

Critical Complements in Fabius Conditioning

Logarithmic overlap, finite-memory corrections,
and the order-zero Rényi crossover

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Abstract

Exponential tilting turns a very small value of a geometric uniform random series into a typical mean constraint. Previous work in the `ProveIt` repository classifies the conditional law of a coordinate block by its share of the tilted variance, but leaves the critical complement regime as an explicit research question. We resolve the fixed-complement case for geometric weights. After observing the active coordinates except for a fixed number r of early digits, the likelihood ratio reduces, eventually exactly, to a density h_r of a tilted boundary tail plus an independent gamma variable. This gives the overlap expansion

$$1 - d_{\text{TV}}(P_{n,r}, Q_{n,r}) = \frac{\Lambda_n + r \log \Lambda_n + \log K - \log(r!) + 1 + o(1)}{\sqrt{2\pi n}}, \quad \Lambda_n = \frac{1}{2} \log(2\pi n).$$

A polynomial tail formula yields an all-orders inverse-logarithmic expansion and an explicit residual endpoint defect. Every fixed positive Rényi order has divergence Λ_n minus the corresponding boundary differential entropy, whereas order zero tends to $\log 2$. We identify the intervening scale $\alpha \sqrt{n} = O(1)$ and its universal normal-tail profile. With no early digit omitted, the remaining overlap correction is positive and has logarithm asymptotic to $-\sqrt{\log(1/q) \log n}$: the flat Fabius endpoint is visible in a configuration-level information statistic. Proofs are self-contained; 228 exact algebra checks and reproducible numerical diagnostics accompany the article.

Status. This is an unrefereed research manuscript with conventional mathematical proofs, not a Lean-verified development. It resolves a precise fixed- r part of a repository research question, not every critical complement regime. Worldwide priority and the description “breakthrough” are not certified. Classical tilting, gamma/simplex identities, and small-deviation estimates are explicitly distinguished from the proposed extensions.

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1 The repository question and the contribution

The Fabius branch of ProveIt studies the compactly supported smooth Fabius–Rvachev functions through exact values, Fourier products, geometric uniform sums, and endpoint asymptotics. These connections are also classical; see Arias de Reyna [2, 3]. The recent repository manuscript *Sharp Conditioning Laws for Uniform Random Series* [1] develops a configuration-level theory rather than only a scalar small-ball estimate.

That manuscript proves, for arbitrary positive summable weights, a variance-fraction criterion for equivalence of a selected conditional block and its exponentially tilted product marginal. At a fixed retained variance fraction strictly below one it obtains Gaussian information profiles. At fraction one it proves asymptotic separation. It also obtains an independent exponential slack, a geometric boundary law, a bridge limit, and the leading full-law overlap. We use this work to identify the question, not as an unproved premise of the arguments below.

The source’s question labelled `q:critical` asks what replaces the Gaussian profile when the complement variance stays bounded, or diverges much more slowly than the total variance. Its separate question `q:edgeworth` concerns phase-sensitive higher corrections. The fixed-complement case has two features that a Gaussian substitution cannot recover: a genuinely non-Gaussian boundary density, and its very flat left endpoint. This article makes both effects explicit.

1.1 What is proved, and what is not

Our retained block is $\{r + 1, \dots, n\}$ for a *fixed* integer $r \geq 0$, along a fixed geometric phase. The omitted part contains r strongly tilted early coordinates and the entire infinite geometric tail. Its tilted variance tends to a positive finite constant, while the retained variance is $n + O(1)$. We obtain a sharp overlap rate, its constant and logarithmic corrections, an effective one-dimensional reduction, all fixed positive information orders, and an order-zero transition. The discarded coordinates and slack also admit an explicit joint approximation in total variation with error $O((\log n)^2/n)$.

The case of a number $r = r_n$ tending to infinity is *not* included in these uniform error claims. Neither are arbitrary nongeometric weights, several simultaneous constraints, an exact random-bit sampler, or a machine-checked formalization. The last section formulates these as further questions instead of treating them as corollaries.

1.2 Relation to existing methods

Exponential-family conditioning and finite-block approximation have a substantial history; Diaconis and Freedman [5] are an important reference point. Small deviations of exponentially weighted independent sums, including the leading logarithm-squared scale used below, belong to an established theory [4]. The information conventions are standard [6]. The proposed contribution here is the particular *critical moving-block theorem package*: its exact boundary kernel, sharp overlap expansion, inverse-logarithmic hierarchy, endpoint-defect transfer, and order-zero crossover. The targeted source review does not establish that no equivalent theorem exists elsewhere.

2 Model, normalization, and the critical block

Fix $q \in (0, 1)$ and $\rho > 0$. All constants in asymptotic bounds may depend on q, ρ , and a fixed r , unless otherwise stated. All logarithms are natural. Under the original probability law P , let $(U_k)_{k \geq 1}$ be independent uniform variables on $[0, 1]$, and define

$$S_q = \sum_{k \geq 1} q^k U_k, \quad t_n = \rho q^{-n}, \quad Y_{n,k} = t_n q^k U_k. \quad (2.1)$$

The series converges absolutely and is bounded by $q/(1-q)$. Write

$$\frac{dQ_n}{dP} = \frac{e^{-t_n S_q}}{\mathbb{E}_P e^{-t_n S_q}}, \quad \mu_n = \mathbb{E}_{Q_n}(t_n S_q), \quad x_n = \mu_n/t_n, \quad (2.2)$$

and let $P_n = P(\cdot \mid S_q \leq x_n)$. We condition the original law, not Q_n , on the inequality. This distinction determines the slack factor and all constants below.

Under Q_n the variables $Y_{n,k}$ remain independent, and the variable with cap $a_{n,k} = \rho q^{k-n}$ has density

$$e_a(y) = \frac{e^{-y}}{1 - e^{-a}} \mathbf{1}_{(0,a)}(y). \quad (2.3)$$

Indeed, the tilt factors over every finite collection; the residual bounded series supplies the independent remaining factor. Its mean and variance, obtained by integrating or differentiating the normalizer, are

$$m(a) = 1 - \frac{a}{e^a - 1}, \quad v(a) = 1 - \frac{a^2 e^{-a}}{(1 - e^{-a})^2}. \quad (2.4)$$

As $a \downarrow 0$, $m(a) = a/2 + O(a^2)$ and $v(a) = a^2/12 + O(a^4)$. As $a \rightarrow \infty$, both tend to one exponentially fast.

Definition 2.1 (Observed block and overlap). For fixed $r \in \mathbb{N} \cup \{0\}$ and $n > r$, put

$$A_{n,r} = \{r+1, \dots, n\}, \quad V_{n,r} = \sum_{k=r+1}^n Y_{n,k}, \quad W_{n,r} = \mu_n - V_{n,r}.$$

Let $P_{n,r}$ and $Q_{n,r}$ be the laws of the vector $(Y_{n,k})_{k \in A_{n,r}}$ under P_n and Q_n , respectively. Set

$$\mathcal{O}_{n,r} = 1 - d_{\text{TV}}(P_{n,r}, Q_{n,r}), \quad c_n = (2\pi n)^{-1/2}, \quad \Lambda_n = -\log c_n. \quad (2.5)$$

Here $d_{\text{TV}}(P, Q) = \sup_B |P(B) - Q(B)|$, or one half of the L^1 distance between densities. Thus $\mathcal{O} = \int \min(p, q)$.

We will reserve the letter E_n for the *actual slack* $\mu_n - \sum_{k \geq 1} Y_{n,k}$ under P_n . The observed deficit $W_{n,r}$ also includes the omitted contributions and is not the same variable.

2.1 The Fabius specialization

For $q = 1/2$, S_q is supported on $[0, 1]$. Its distribution function $F(x) = P(S_q \leq x)$ obeys $S_q \stackrel{d}{=} (U + S'_q)/2$, with independence. Differentiating the resulting convolution gives

$$F'(x) = 2\{F(2x) - F(2x-1)\},$$

where $F = 0$ to the left of zero and $F = 1$ to the right of one. In particular $F'(x) = 2F(2x)$ for $0 < x < 1/2$. This is the probability normalization of the Fabius function used in the geometric-sum representation [2]. Our event is therefore a Fabius endpoint event when $q = 1/2$, not an unrelated finite-dimensional surrogate.

3 The boundary kernel and the main results

Introduce an *untitled* copy of the scaled geometric tail,

$$T_\rho = \rho \sum_{j \geq 1} q^j U_j, \quad C = \frac{\rho q}{1-q}, \quad K = \frac{1}{\mathbb{E} e^{-T_\rho}} = \prod_{j \geq 1} \frac{\rho q^j}{1 - e^{-\rho q^j}}. \quad (3.1)$$

Let R_ρ have the exponentially tilted law of T_ρ :

$$\mathcal{L}(R_\rho)(dz) = K e^{-z} \mathcal{L}(T_\rho)(dz). \quad (3.2)$$

Equivalently, R_ρ is a sum of independent variables of density (2.3) with caps ρq^j , $j \geq 1$. It is exactly the Q_n -law of $\sum_{k>n} Y_{n,k}$, for every n .

Let $G_{r+1} \sim \text{Gamma}(r+1, 1)$ be independent of R_ρ . The shape parameter is $r+1$, not r : the extra exponential is the one-sided slack. Define

$$H_r = R_\rho + G_{r+1}, \quad h_r(w) = \frac{K e^{-w}}{r!} \mathbb{E}(w - T_\rho)_+^r \quad (w > 0). \quad (3.3)$$

For $r = 0$, $(w - T)_+^0$ in this formula means $\mathbf{1}_{\{T < w\}}$; thus $h_0(w) = K e^{-w} P(T_\rho < w)$. The value at an endpoint is immaterial. We prove in section 4 that h_r is the probability density of H_r .

Theorem 3.1 (Sharp critical overlap). *For each fixed q, ρ, r as above, as $n \rightarrow \infty$,*

$$\mathcal{O}_{n,r} = c_n \{ \Lambda_n + r \log \Lambda_n + \log K - \log(r!) + 1 + o(1) \}. \quad (3.4)$$

More precisely, with

$$J_r(z) = \int_0^\infty \min\{1, h_r(w)/z\} dw, \quad (3.5)$$

one has the quantitative reduction

$$\frac{\mathcal{O}_{n,r}}{c_n} = J_r(c_n) + O\left(\frac{(\log n)^3}{n}\right). \quad (3.6)$$

The likelihood ratio of the full observed vector depends only on $V_{n,r}$. Consequently all the divergences stated in this article are also the divergences of the corresponding one-dimensional sum laws.

The leading coefficient of $\log n / \sqrt{n}$ is the same for every fixed r . Omitting one additional early digit changes the next scale by $\log \log n / \sqrt{n}$. The phase parameter enters already through $\log K / \sqrt{n}$. Equal-prior optimal binary testing error is precisely $\mathcal{O}_{n,r}/2$, so the expansion has an immediate statistical meaning.

For a density h , write its differential Rényi entropies as

$$\mathfrak{h}_\alpha(h) = \frac{\log \int h^\alpha}{1-\alpha} \quad (0 < \alpha < \infty, \alpha \neq 1), \quad \mathfrak{h}_1(h) = - \int h \log h, \quad \mathfrak{h}_\infty(h) = - \log \|h\|_\infty. \quad (3.7)$$

For probability measures, $D_\alpha(P\|Q)$ denotes the usual Rényi divergence, with the continuous order-one and essential-supremum order-infinity conventions [6].

Theorem 3.2 (Boundary information constants). *For every fixed $\alpha \in (0, \infty]$,*

$$D_\alpha(P_{n,r}\|Q_{n,r}) = \Lambda_n - \mathfrak{h}_\alpha(h_r) + o(1). \quad (3.8)$$

This includes $D_{\text{KL}}(P_{n,r}\|Q_{n,r}) = \Lambda_n - \mathfrak{h}_1(h_r) + o(1)$. Under P_n , $W_{n,r}$ converges to H_r in total variation with error $O((\log n)^2/n)$. The reverse relative entropy $D_{\text{KL}}(Q_{n,r}\|P_{n,r})$ is infinite for every sufficiently large n .

Theorem 3.3 (The order-zero transition). *At order zero,*

$$D_0(P_{n,r}\|Q_{n,r}) \longrightarrow \log 2. \quad (3.9)$$

More generally, if $\alpha_n \geq 0$ and $\alpha_n \sqrt{n} \rightarrow c \in [0, \infty)$, then

$$D_{\alpha_n}(P_{n,r}\|Q_{n,r}) \longrightarrow C(c) := - \log \left(e^{c^2/2} \Phi(-c) \right), \quad (3.10)$$

where Φ is the standard normal distribution function. The limiting function does not depend on r, q , or ρ .

Theorem 3.4 (The flat-endpoint correction). *Suppose $r = 0$ and put $\lambda = \log(1/q)$. Define*

$$\delta_0(z) = \int_0^C \left(1 - \frac{h_0(w)}{z}\right)_+ dw.$$

For sufficiently large n ,

$$\frac{\mathcal{O}_{n,0}}{c_n} = \Lambda_n + \log K + 1 - \delta_0(c_n) + O\left(\frac{(\log n)^3}{n}\right), \quad \frac{\log \delta_0(c_n)}{\sqrt{\log n}} \rightarrow -\sqrt{\lambda}. \quad (3.11)$$

In particular the residual

$$\mathcal{E}_n = \Lambda_n + \log K + 1 - \frac{\mathcal{O}_{n,0}}{c_n}$$

is eventually strictly positive and satisfies

$$\frac{\log \mathcal{E}_n}{\sqrt{\log n}} \rightarrow -\sqrt{\log(1/q)}. \quad (3.12)$$

This is a statement about the *scaled* overlap residual. In the unscaled overlap the correction is $c_n \delta_0(c_n)$. It is smaller than $c_n(\log n)^{-M}$ for every fixed M , but larger than the $O(c_n(\log n)^3/n)$ bulk error. No multiplicative equivalent for δ_0 is asserted.

4 Exact boundary algebra and the variance clock

Lemma 4.1 (Kernel formula and regularity). *The product K in (3.1) converges to a finite positive number, and (3.3) is a density. It is continuous, positive for $w > 0$, zero for $w \leq 0$, bounded by one, and has an exponentially decreasing polynomial right tail. For every $\alpha > 0$, $\int h_r^\alpha < \infty$ and $\int |h_r \log h_r| < \infty$.*

Proof. For small a , $\log(a/(1 - e^{-a})) = a/2 + O(a^2)$; summability of ρq^j proves product convergence. The finite-coordinate Laplace products tend to the infinite product by bounded convergence. Tilting T_ρ then gives (3.2); in particular

$$\mathbb{E}[e^{R_\rho} g(R_\rho)] = K \mathbb{E}[g(T_\rho)]$$

for bounded measurable g . Convolve R_ρ with the gamma density $e^{-s} s^r / r!$ on $s > 0$. The convolution equals

$$\frac{e^{-w}}{r!} \mathbb{E}[e^{R_\rho} (w - R_\rho)^r \mathbf{1}_{\{R_\rho < w\}}],$$

which is (3.3). Hence its integral is one. The gamma density is bounded by one (it is a convolution of $r + 1$ rate-one exponential densities), so the same is true after convolving with R_ρ .

The original tail T_ρ has no atoms because its first uniform summand has a density; it takes values arbitrarily near zero with positive probability. This gives continuity and strict positivity of h_r on $(0, \infty)$, including the $r = 0$ case. Since $T_\rho \leq C$, (3.3) is bounded for large w by a constant times $e^{-w}(1 + w)^r$. Every positive power is therefore integrable. Finally $y |\log y| \leq C_0 \sqrt{y}$ on $0 < y \leq 1$, which proves the entropy integrability assertion. $\square$

Let $m_j^* = \mathbb{E} T_\rho^j$, with $m_0^* = 1$, and define the monic polynomial

$$p_r(w) = \mathbb{E}(w - T_\rho)^r = \sum_{j=0}^r (-1)^j \binom{r}{j} m_j^* w^{r-j}. \quad (4.1)$$

Then for $w \geq C$,

$$h_r(w) = \frac{K}{r!} e^{-w} p_r(w), \quad p_r'(w) = r p_{r-1}(w). \quad (4.2)$$

The entire infinite boundary enters this tail through finitely many ordinary moments and the scalar K . It is important that these are the moments of *untilded* T_ρ , not of R_ρ .

Proposition 4.2 (Exact moment recursion). *The moments satisfy*

$$m_k^* = \frac{1}{1-q^k} \sum_{j=0}^{k-1} \binom{k}{j} \frac{(\rho q)^{k-j} q^j}{k-j+1} m_j^*, \quad k \geq 1. \quad (4.3)$$

Thus m_k^* is rational whenever q and ρ are rational. The kernels also satisfy

$$\left(\frac{d}{dw} + 1 \right) h_{r+1}(w) = h_r(w), \quad w > 0. \quad (4.4)$$

Proof. The identity $T_\rho \stackrel{d}{=} \rho q U + q T'_\rho$ has independent terms. Taking its k th moment, the term $j = k$ on the right is $q^k m_k^*$; isolating it gives (4.3). The uniform moment is $\mathbb{E} U^{k-j} = 1/(k-j+1)$. Differentiating the integrated positive-part formula in (3.3) proves (4.4); for $r = 0$ this uses the continuous distribution function of T_ρ . Equivalently, $h_{r+1} = h_r * e^{-w} \mathbf{1}_{w>0}$, which gives the same identity. $\square$

Proposition 4.3 (Mean, variance, and phase dependence). *Write $\mu_R = \mathbb{E} R_\rho$ and $\sigma_R^2 = \text{Var } R_\rho$. Then*

$$\mu_n = n + \kappa + o(1), \quad \kappa = \mu_R - \sum_{j \geq 0} \frac{\rho q^{-j}}{e^{\rho q^{-j}} - 1}, \quad (4.5)$$

$$\text{Var}_{Q_n} \left(\sum_{k \geq 1} Y_{n,k} \right) = n + \nu + o(1), \quad \nu = \sigma_R^2 - \sum_{j \geq 0} \{1 - v(\rho q^{-j})\}. \quad (4.6)$$

For fixed r the omitted variance is $r + \sigma_R^2 + o(1)$, and

$$\frac{\text{Var}_{Q_n} V_{n,r}}{\text{Var}_{Q_n} (\sum_{k \geq 1} Y_{n,k})} = 1 - \frac{r + \sigma_R^2}{n} + o(n^{-1}). \quad (4.7)$$

Furthermore

$$\frac{d}{d\rho} \log K = \frac{\mu_R}{\rho} > 0. \quad (4.8)$$

Proof. Reindex the first n caps as ρq^{-j} , $0 \leq j < n$. Their mean sum is $n - \sum_{j=0}^{n-1} (1 - m(\rho q^{-j}))$; the remaining mean is μ_R . The deficits decrease superexponentially in j , so the sum converges and its omitted part tends to zero. Apply the same argument to variances. The first r caps tend to infinity, so their variances tend to one. Independence gives (4.7). Finally differentiate the convergent product locally uniformly in $\rho > 0$. The derivative of its j th logarithm is $m(\rho q^j)/\rho$; summing proves (4.8). $\square$

One may restrict ρ to $[1, q^{-1})$ to use a canonical multiplicative phase. None of the proofs needs that restriction. No uniformity as $q \rightarrow 1$, $\rho \rightarrow 0$, or $r \rightarrow \infty$ is implicit.

5 An eventually exact likelihood-ratio formula

Let $f_{n,r}$ be the density of $V_{n,r}$ under Q_n , extended by zero outside its support. The next identity is the reason the problem remains tractable even though the retained vector has growing dimension.

Theorem 5.1 (Exact critical kernel). *For every sufficiently large n , define*

$$Z_{n,r} = \int_0^\infty f_{n,r}(\mu_n - w) h_r(w) dw. \quad (5.1)$$

Then

$$\frac{dP_{n,r}}{dQ_{n,r}}(y) = \frac{h_r(\mu_n - \sum_{k \in A_{n,r}} y_k)}{Z_{n,r}}. \quad (5.2)$$

Under P_n , the density of the observed deficit $W_{n,r}$ is

$$g_{n,r}(w) = \frac{f_{n,r}(\mu_n - w) h_r(w)}{Z_{n,r}}. \quad (5.3)$$

Proof. Set $B_n = \mathbb{E}_{Q_n}[e^{T_n - \mu_n} \mathbf{1}_{\{T_n \leq \mu_n\}}]$, where $T_n = \sum_{k \geq 1} Y_{n,k}$. Undoing the tilt in (2.2) gives

$$\frac{dP_n}{dQ_n} = \frac{e^{T_n - \mu_n} \mathbf{1}_{\{T_n \leq \mu_n\}}}{B_n}.$$

Split $T_n = V_{n,r} + L_n^{\text{lost}}$, with the two parts independent under Q_n . Integrating out the lost part, and writing $w = \mu_n - v$, gives the marginal numerator

$$e^{-w} \mathbb{E}_{Q_n}[e^{L_n^{\text{lost}}} \mathbf{1}_{\{L_n^{\text{lost}} \leq w\}}]. \quad (5.4)$$

This is the density at w of $L_n^{\text{lost}} + E$, where $E \sim \text{Exp}(1)$ is independent.

The lost part consists of the first r coordinates and a tail with law R_ρ . If $r > 0$, its smallest early cap is $a_{n,r} = \rho q^{r-n}$, and $\mu_n < a_{n,r}$ eventually by theorem 4.3. On the domain in (5.4), where $0 \leq w \leq \mu_n$, every early lost coordinate is below every early cap. Their joint density is therefore the joint density of r independent rate-one exponentials multiplied by the constant

$$p_{n,r}^{-1}, \quad p_{n,r} = \prod_{k=1}^r (1 - e^{-a_{n,k}}).$$

For $r = 0$, set $p_{n,0} = 1$. Thus (5.4) equals $h_r(w)/p_{n,r}$ throughout the only region accessible to the nonnegative observed sum. The same constant occurs in B_n , so it cancels. In fact $Z_{n,r} = p_{n,r} B_n$. This proves (5.2); changing variables from v to w proves (5.3). $\square$

Remark 5.2 (Which normalizer is being used). $Z_{n,r}$ is the normalizer for the fixed limiting kernel h_r . For $r > 0$ it is not literally the original rejection normalizer B_n , although $p_{n,r} \rightarrow 1$ extremely fast. Keeping the exact factor $p_{n,r}$ prevents a finite- n normalization error.

Corollary 5.3 (Sufficiency of the sum). *Every integral divergence obtained by integrating a function of $dP_{n,r}/dQ_{n,r}$ is unchanged when the observed vector is replaced by its sum $V_{n,r}$. This includes overlap, total variation, relative entropy, and every Rényi order used here.*

Proof. The likelihood ratio in (5.2) is a measurable function of the sum. The pushforward likelihood ratio is the same function, since integration over each sum preimage leaves it unchanged. The two integrals are then equal by the definition of pushforward measure. The essential supremum is likewise unchanged. $\square$

6 A local density estimate with an elementary proof

We need a local approximation on a logarithmic window, not a central limit theorem alone. In this geometric setting it can be proved directly from a large gamma core and a small boundary block.

Lemma 6.1 (Gamma-core local estimate). *For fixed q, ρ, r , there is a constant C_0 such that*

$$\sup_v f_{n,r}(v) \leq C_0 n^{-1/2} \quad (6.1)$$

for sufficiently large n . For every fixed $B > 0$,

$$\sup_{|u| \leq B \log n} \left| \frac{f_{n,r}(\mathbb{E}_{Q_n} V_{n,r} + u)}{c_n} - 1 \right| = O_B \left(\frac{(\log n)^2}{n} \right). \quad (6.2)$$

Also

$$\frac{V_{n,r} - \mathbb{E}_{Q_n} V_{n,r}}{\sqrt{n}} \Rightarrow N(0, 1). \quad (6.3)$$

Proof. Choose a large fixed M , say $M > 10$. Take $\ell_n = O(\log \log n)$ so that the smallest cap in the core $\{r+1, \dots, n - \ell_n\}$ is at least $M \log n$; the remaining ℓ_n coordinates form a boundary block. Denote the core size by $N = n - r - \ell_n$.

Step 1: removing core caps. The core law is exactly a vector of N independent exponentials conditioned to stay below its coordinate caps. If the smallest cap is a , a union bound bounds the probability of a violation by

$$\sum_{j \geq 0} e^{-aq^{-j}} \leq \frac{e^{-a}}{1 - e^{-a(q^{-1}-1)}} = O(n^{-M}).$$

Here $q^{-j} \geq 1 + j(q^{-1} - 1)$. Hence the actual core sum and a $\text{Gamma}(N, 1)$ sum differ in total variation by $O(n^{-M})$. Their means differ by at most $\sum a_j/(e^{a_j} - 1) = O((\log n)n^{-M})$.

Let D_n be the sum of the boundary block, independent of either core construction. It contains the last coordinate, whose density is bounded by $(1 - e^{-\rho})^{-1}$. Consequently its density has the same upper bound. Convolution upgrades the total-variation error of the core to a uniform density error $O(n^{-M})$ for the full sum: for measures η, η' and a bounded density d ,

$$\sup_x |((\eta - \eta') * d)(x)| \leq 2\|d\|_\infty d_{\text{TV}}(\eta, \eta').$$

Thus it suffices to analyze $G_N + D_n$, where $G_N \sim \text{Gamma}(N, 1)$.

Step 2: size of the boundary block. A capped exponential is stochastically dominated by an uncapped rate-one exponential. This follows, for example, by coupling their inverse distribution functions:

$$-\log\{1 - (1 - e^{-a})U\} \leq -\log(1 - U).$$

Therefore $\mathbb{E}D_n \leq \ell_n$ and $\mathbb{E}e^{D_n/2} \leq 2^{\ell_n}$. For every prescribed $A > 0$, choosing a fixed sufficiently large M' gives

$$P(D_n > M' \log n) \leq 2^{\ell_n} n^{-M'/2} = O(n^{-A}). \quad (6.4)$$

Step 3: the gamma density near its mean. Write $\gamma_N(x) = e^{-x} x^{N-1} / \Gamma(N)$ for $x > 0$. Stirling's formula with its $O(N^{-1})$ logarithmic remainder, followed by the Taylor expansion of $\log(1 + s/N)$, gives

$$\gamma_N(N + s) = \frac{1}{\sqrt{2\pi N}} \left\{ 1 + O\left(\frac{1 + s^2}{N}\right) \right\} \quad \text{uniformly for } |s| \leq C \log n. \quad (6.5)$$

For clarity, the logarithm of the density minus $-\frac{1}{2} \log(2\pi N)$ is $-s/N - s^2/(2N) + O(N^{-1} + s^2/N^2 + |s|^3/N^2)$ on this window. This tends to zero uniformly and implies (6.5). The gamma density has its maximum at $N - 1$ when $N > 1$, so Stirling also gives $\|\gamma_N\|_\infty \leq C_1/\sqrt{N}$.

At $v = \mathbb{E}V_{n,r} + u$ with $|u| \leq B \log n$, the argument $v - D_n$ is $N + O(\log n)$ on the event in (6.4). Integrate (6.5) over that event. Its complement is bounded using the global gamma maximum and (6.4). Replacing N by n changes the leading density by $1 + O(\log \log n/n)$, within the stated error. The uniform $O(n^{-M})$ density error from Step 1 is smaller still after division by c_n . This proves (6.2); the global maximum bound and the convolution error prove (6.1).

Step 4: the central limit theorem. The centered gamma core is the sum of N centered independent exponentials. Its characteristic function at $u/\sqrt{N}$ is

$$\left(\frac{e^{-iu/\sqrt{N}}}{1 - iu/\sqrt{N}} \right)^N \rightarrow e^{-u^2/2},$$

by expanding the logarithm. Hence it converges to the standard normal law. The centered boundary block divided by $\sqrt{n}$ tends to zero in probability, since $\mathbb{E}|D_n - \mathbb{E}D_n| \leq 2\ell_n = o(\sqrt{n})$. The core total-variation error tends to zero and $N/n \rightarrow 1$. These facts yield (6.3). $\square$

Corollary 6.2 (Normalizer and local flatness at the constraint). *For fixed $B > 0$,*

$$Z_{n,r} = c_n \{1 + O((\log n)^2/n)\}, \quad (6.6)$$

$$\frac{f_{n,r}(\mu_n - w)}{c_n} = 1 + O_B((\log n)^2/n), \quad 0 \leq w \leq B \log n. \quad (6.7)$$

Moreover $f_{n,r}(\mu_n - w)/c_n$ is bounded uniformly in w, n .

Proof. The difference $\mu_n - \mathbb{E}V_{n,r}$ is the omitted mean, which tends to $r + \mu_R$ and is bounded. Thus (6.7) follows from [theorem 6.1](#). Integrate it against h_r on $[0, B \log n]$. The omitted tail has mass $O(n^{-B}(\log n)^r)$ by (4.2); the global bound controls its contribution. Taking B sufficiently large proves (6.6). $\square$

Corollary 6.3 (A joint critical-complement approximation). *Under P_n , the joint law of the first r rescaled digits, the entire rescaled tail after n , and the actual slack E_n is within $O((\log n)^2/n)$ in total variation of*

$$\text{Exp}(1)^{\otimes r} \otimes \bigotimes_{j \geq 1} e_{pq^j}(z_j) dz_j \otimes \text{Exp}(1). \quad (6.8)$$

In particular $W_{n,r}$ is within the same order of total variation of H_r .

Proof. The exact cancellation in [theorem 5.1](#), now retaining all lost coordinates and the slack, gives the joint likelihood ratio

$$\frac{f_{n,r}(\mu_n - w)}{Z_{n,r}}, \quad w = \sum_{i=1}^r e_i + \sum_{j \geq 1} z_j + e,$$

relative to the product law in (6.8). This can also be checked directly: transforming the observed sum to the slack multiplies its density by e^{-e} after undoing the tilt; the early lost coordinate densities are constant multiples of e^{-e_i} , and those constants cancel the normalizer as before. Under the product reference, w has density h_r .

On $w \leq B \log n$, [theorem 6.2](#) bounds the likelihood ratio minus one by $O((\log n)^2/n)$. On the complement the ratio is uniformly bounded, and the reference mass is $O(n^{-B}(\log n)^r)$. Integrating the absolute difference proves the assertion. Taking the sum is a measurable map and cannot increase total variation. $\square$

The qualitative independence of boundary coordinates and slack is already part of the preceding repository work. The quantitative fixed-complement estimate above is included to make the later information limits independent of any unproved uniform local theorem.

7 Overlap: a logarithmic threshold and an exact tail formula

7.1 Reduction to a fixed density

Proof of (3.6). By [theorem 5.1](#),

$$\mathcal{O}_{n,r} = \int_0^\infty f_{n,r}(\mu_n - w) \min\{1, h_r(w)/Z_{n,r}\} dw. \quad (7.1)$$

Put $\varepsilon_n = (\log n)^2/n$ and choose a sufficiently large fixed B . On $[0, B \log n]$, replacing $f_{n,r}/c_n$ by one introduces an error $O(\varepsilon_n \log n)$. Replacing $Z_{n,r}$ by c_n has the same cost: if $Z/c = 1 + O(\varepsilon_n)$, then for every $h \geq 0$,

$$|\min(1, h/Z) - \min(1, h/c)| \leq C_2 \varepsilon_n \min(1, h/c).$$

The integral of the last minimum is $O(\log n)$, since it is at most $B \log n$ on the window, and at most $c_n^{-1} \int_{B \log n}^\infty h_r$ outside it. The tails in (7.1), after division by c_n , have this same upper bound times a fixed constant. They are $O(n^{1/2-B}(\log n)^r)$, negligible compared with $(\log n)^3/n$ when B is chosen large. This proves the reduction. $\square$

7.2 A finite polynomial controls the upper threshold

Put $A = K/r!$, and choose the fixed cutoff $D = C + r + 1$. The symbol D in this subsection is a cutoff, not a divergence. For sufficiently small $z > 0$, let $b_r(z) > D$ be the unique solution of

$$Ae^{-b_r(z)}p_r(b_r(z)) = z, \quad (7.2)$$

and define the left endpoint defect

$$\delta_r(z) = \int_0^D \left(1 - \frac{h_r(w)}{z}\right)_+ dw. \quad (7.3)$$

The value of this integral is unchanged by enlarging the fixed cutoff once z is sufficiently small.

Proposition 7.1 (Exact decomposition of the overlap kernel). *For all sufficiently small $z > 0$,*

$$J_r(z) = b_r(z) + \frac{\sum_{j=0}^r p_r^{(j)}(b_r(z))}{p_r(b_r(z))} - \delta_r(z). \quad (7.4)$$

In particular $\delta_r(z) \rightarrow 0$ as $z \downarrow 0$.

Proof. For $w > C$,

$$\frac{p_r'(w)}{p_r(w)} = \frac{r\mathbb{E}(w - T_\rho)^{r-1}}{\mathbb{E}(w - T_\rho)^r} \leq \frac{r}{w - C},$$

with zero numerator when $r = 0$. Thus $e^{-w}p_r(w)$ is strictly decreasing on $[D, \infty)$ and tends to zero. If $z < h_r(D)$, the root (7.2) exists and is unique. Repeated integration by parts, which terminates after r steps, gives

$$\int_b^\infty e^{-w}p_r(w)dw = e^{-b} \sum_{j=0}^r p_r^{(j)}(b).$$

The integral defining J_r is $D - \delta_r(z)$ on $[0, D]$, is $b_r(z) - D$ on $[D, b_r(z)]$, and is the displayed tail integral multiplied by A/z thereafter. Equation (7.2) cancels the last exponential and proves (7.4). Finally $h_r(w) > 0$ for every $w > 0$, so dominated convergence on $[0, D]$ gives $\delta_r(z) \rightarrow 0$. $\square$

Proof of theorem 3.1. Set $z = c_n$ and abbreviate $b = b_r(c_n)$, $\Lambda = \Lambda_n$. Taking logarithms in (7.2), and using $p_r(b) = b^r(1 + O(b^{-1}))$, gives

$$b = \Lambda + \log A + r \log b + O(b^{-1}).$$

First this implies $b/\Lambda \rightarrow 1$; substituting back then gives $b = \Lambda + r \log \Lambda + \log A + o(1)$. The rational tail factor in (7.4) tends to one, and $\delta_r(c_n) \rightarrow 0$. Combine with (3.6) and $\log A = \log K - \log(r!)$. The sufficiency assertion follows from theorem 5.3. $\square$

7.3 The inverse-logarithmic hierarchy

The exact decomposition is stronger than the three-term formula. For the next statement put

$$m_1^* = \frac{\rho q}{2(1-q)}, \quad d_n = r \log \Lambda_n + \log A.$$

Proposition 7.2 (First higher correction and all orders). *For fixed r , one has*

$$\begin{aligned} \frac{\mathcal{O}_{n,r}}{c_n} &= \Lambda_n + d_n + 1 + \frac{rd_n + r(1 - m_1^*)}{\Lambda_n} - \delta_r(c_n) \\ &\quad + O\left(\frac{(\log \Lambda_n)^2}{\Lambda_n^2} + \frac{(\log n)^3}{n}\right). \end{aligned} \quad (7.5)$$

More generally, for every fixed integer $M \geq 1$ there are explicitly recursive polynomials $A_1(X), \dots, A_M(X)$, with $\deg A_j \leq j$, such that

$$\begin{aligned} \frac{\mathcal{O}_{n,r}}{c_n} &= \Lambda_n + r \log \Lambda_n + \log A + 1 + \sum_{j=1}^M \frac{A_j(\log \Lambda_n)}{\Lambda_n^j} - \delta_r(c_n) \\ &\quad + O\left(\frac{(\log \Lambda_n)^{M+1}}{\Lambda_n^{M+1}} + \frac{(\log n)^3}{n}\right). \end{aligned} \quad (7.6)$$

Their coefficients depend only on $r, K, m_1^*, \dots, m_r^*$. In particular $A_1(X) = r^2 X + r \log A + r(1 - m_1^*)$. For $r = 0$, all these polynomials vanish and the upper-tail part of (7.4) is exactly $\Lambda_n + \log K + 1$.

Proof. For large b ,

$$\log p_r(b) = r \log b - \frac{rm_1^*}{b} + O(b^{-2}).$$

Writing $b = \Lambda + d + \eta$ in the logarithmic root equation and expanding $\log b$ yields

$$b = \Lambda + d + \frac{rd - rm_1^*}{\Lambda} + O((\log \Lambda)^2 / \Lambda^2).$$

The rational tail factor in (7.4) is $1 + r/b + O(b^{-2})$. Combining these proves (7.5).

For completeness the same procedure has controlled remainders at every fixed order. Write $p_r(b) = b^r(1 + a_1/b + \dots + a_r/b^r)$ and consider

$$F(b) = b - r \log b - \log(1 + a_1/b + \dots + a_r/b^r).$$

For sufficiently large b , $F'(b) = 1 - r/b + O(b^{-2}) > 1/2$. Starting with $b_0 = \Lambda + r \log \Lambda + \log A$, substitute

$$b_M = b_0 + \sum_{j=1}^M B_j(\log \Lambda) \Lambda^{-j}$$

into $F(b_M) = \Lambda + \log A$. Taylor-expand the logarithms in Λ^{-1} . At order Λ^{-j} the new polynomial B_j appears with coefficient one; all other terms are already determined. Set B_j equal to their negative. Products of the previous terms show inductively that $\deg B_j \leq j$. Taylor's theorem gives a residual $O((\log \Lambda)^{M+1} / \Lambda^{M+1})$. Since $F' > 1/2$, the mean value theorem bounds $|b - b_M|$ by twice this residual.

The rational function $\sum p_r^{(j)} / p_r$ has a convergent power series in $1/b$ near infinity. Substitute the same b_M and expand to order M ; its bounded derivative transfers the root remainder. This gives the polynomials A_j and the stated error for J_r . Equation (3.6) adds the bulk error. When $r = 0$, $p_0 = 1$, $b = \Lambda + \log K$, and the tail factor is exactly one. $\square$

Section 10 proves that $\delta_r(c_n)$ is smaller than every fixed inverse power of Λ_n . Thus (7.6) is also an ordinary all-orders inverse-logarithmic expansion after suppressing the defect. Retaining the defect is substantially more informative at accessible values of n , and is essential when $r = 0$.

8 Information constants and the non-Gaussian deficit

Proof of theorem 3.2. The total-variation statement is theorem 6.3. For $0 < \alpha < \infty$, $\alpha \neq 1$, the exact likelihood formula gives

$$\exp\{(\alpha - 1)D_\alpha(P_{n,r} \| Q_{n,r})\} = Z_{n,r}^{-\alpha} \int_0^\infty f_{n,r}(\mu_n - w) h_r(w)^\alpha dw. \quad (8.1)$$

For every fixed $w > 0$, $f_{n,r}(\mu_n - w)/c_n \rightarrow 1$, and the ratio is uniformly bounded by [theorem 6.2](#). Since h_r^α is integrable, dominated convergence implies

$$\int_0^\infty f_{n,r}(\mu_n - w) h_r(w)^\alpha dw = c_n \left\{ \int_0^\infty h_r(w)^\alpha dw + o(1) \right\}.$$

Combine this with $Z_{n,r}/c_n \rightarrow 1$ in [\(8.1\)](#). Taking the logarithm and dividing by $\alpha - 1$ proves [\(3.8\)](#) for these orders.

For order one, [\(5.3\)](#) gives the exact identity

$$D_{\text{KL}}(P_{n,r} \| Q_{n,r}) = -\log Z_{n,r} + \frac{1}{Z_{n,r}} \int_0^\infty f_{n,r}(\mu_n - w) h_r(w) \log h_r(w) dw. \quad (8.2)$$

The same argument now uses the integrability of $|h_r \log h_r|$ from [theorem 4.1](#). This proves the Shannon-entropy constant.

The continuous density h_r tends to zero at zero and infinity, so it attains a positive maximum at some finite $w > 0$. For large n , this point lies in the interior of the accessible range of $\mu_n - V_{n,r}$. Indeed $\mu_n \sim n$, whereas the observed sum's support has an exponentially large upper endpoint. The density $f_{n,r}$ is strictly positive in its support interior. Consequently the essential supremum of the likelihood ratio is exactly $\|h_r\|_\infty / Z_{n,r}$, proving the order-infinity formula.

Finally $P_{n,r}(V_{n,r} > \mu_n) = 0$, whereas the same event has positive $Q_{n,r}$ probability for large n (in fact tending to one half by [\(6.3\)](#)). Thus $Q_{n,r}$ is not absolutely continuous with respect to $P_{n,r}$, and the reverse relative entropy is infinite. $\square$

Remark 8.1 (Rates for fixed information orders). The proof also gives an $O_\alpha((\log n)^2/n)$ remainder in [\(3.8\)](#) for each *fixed* positive order, including one and infinity. To see this, use [\(6.7\)](#) on $[0, B \log n]$ and the exponential-polynomial tail of h_r^α outside it, choosing B large depending on α . For order one use $h_r |\log h_r|$ instead. The positive integral $\int h_r^\alpha$ is fixed, so taking its logarithm preserves the error order. The resulting constants are not uniform as $\alpha \downarrow 0$. The theorem in the next section shows why such a qualification is essential.

8.1 Hellinger affinity and the meaning of the constant

The Hellinger affinity has the particularly transparent form

$$\int \sqrt{dP_{n,r} dQ_{n,r}} = (2\pi n)^{-1/4} \int_0^\infty \sqrt{h_r(w)} dw (1 + o(1)) = (2\pi n)^{-1/4} e^{h_{1/2}(h_r)/2} (1 + o(1)). \quad (8.3)$$

This follows either directly from [\(8.1\)](#) or by setting $\alpha = 1/2$ in [theorem 3.2](#).

The retained vector is high-dimensional, but its conditional likelihood is concentrated in an $O(1)$ deficit window, whose profile is h_r . The canonical sum density is approximately c_n on that window. The factor $-\log c_n = \Lambda_n$ therefore measures the large-scale concentration cost, while $h_\alpha(h_r)$ measures the remaining microscopic spread. The overlap samples a larger window, of width $\Lambda_n + r \log \Lambda_n + O(1)$, because its integrand is clipped at one. This distinction explains why the overlap contains a logarithmic factor that is absent from the affinity.

All fixed nonnegative moments of $W_{n,r}$ also converge to those of H_r . For any fixed moment order the error is $O((\log n)^2/n)$: multiply the argument in [theorem 6.3](#) by w^k , use the uniform likelihood bound, and choose a sufficiently large logarithmic cutoff. This moment statement is separate from entropy convergence, which required the logarithmic integrability check above.

9 A universal transition at Rényi order zero

At order zero the divergence records only the Q -mass of the support of P . A narrow high-density region can have a large positive-order divergence while still having a support of macroscopic Q -mass. This distinction is realized sharply here.

Proof of theorem 3.3. Since $h_r(w) > 0$ exactly for $w > 0$, the support-mass formula is

$$D_0(P_{n,r} \| Q_{n,r}) = -\log Q_n(V_{n,r} < \mu_n). \quad (9.1)$$

By theorem 6.1 and the bounded difference $\mu_n - \mathbb{E}V_{n,r}$, the standardized deficit $W_{n,r}/\sqrt{n}$ converges under Q_n to the standard normal law. This proves (3.9).

Now let $\alpha_n > 0$ with $\alpha_n \sqrt{n} \rightarrow c < \infty$; zero orders occurring along a subsequence are already covered. For fixed $0 < \eta < M < \infty$, the polynomial tail formula implies uniformly for $z \in [\eta, M]$,

$$\log h_r(\sqrt{n}z) = -\sqrt{n}z + r \log(\sqrt{n}z) + \log(K/r!) + o(1). \quad (9.2)$$

Multiplication by α_n therefore gives $h_r(\sqrt{n}z)^{\alpha_n} \rightarrow e^{-cz}$ uniformly on that interval. Also $0 \leq h_r^{\alpha_n} \leq 1$ by theorem 4.1.

It follows that

$$\mathbb{E}_{Q_n}[h_r(W_{n,r})^{\alpha_n} \mathbf{1}_{\{W_{n,r} > 0\}}] \rightarrow \int_0^\infty e^{-cz} \phi(z) dz. \quad (9.3)$$

Here is an explicit justification at the two omitted boundaries. The contribution of $0 < W_{n,r} < \eta\sqrt{n}$ is at most its probability, whose upper limit is $\Phi(\eta) - 1/2$. The contribution of $W_{n,r} > M\sqrt{n}$ is at most a probability whose upper limit is $1 - \Phi(M)$. The middle interval follows from weak convergence and uniform convergence of the bounded integrand. Let $\eta \downarrow 0$ and $M \rightarrow \infty$.

Since $\log Z_{n,r} = -\Lambda_n + o(1)$, $Z_{n,r}^{-\alpha_n} \rightarrow 1$. Completing the square evaluates the integral in (9.3) as

$$\int_0^\infty e^{-cz} \phi(z) dz = e^{c^2/2} \Phi(-c) > 0.$$

Use (8.1), take logarithms, and note that $\alpha_n - 1 \rightarrow -1$. This yields (3.10). $\square$

The crossover starts at $C(0) = \log 2$. As a property of the *limiting function*,

$$C(c) = \log(c\sqrt{2\pi}) + O(c^{-2}) \quad (c \rightarrow \infty). \quad (9.4)$$

For example this follows by substituting $u = cz$ in its defining integral and expanding $e^{-u^2/(2c^2)}$ under the integrable weight e^{-u} . This agrees with the formal small-order matching $\Lambda_n + \log \alpha_n = \log(c\sqrt{2\pi})$. It does *not* prove a uniform theorem in a regime where $\alpha_n \sqrt{n} \rightarrow \infty$; such a matched estimate is a further research problem.

10 Flatness survives in the overlap residual

The leading small-deviation scale of an exponentially weighted series is classical [4]. We give the elementary estimate needed here, including its error scale, and then transfer it to the information-theoretic defect.

Lemma 10.1 (Geometric small-ball estimate). *Let $F_\rho(w) = P(T_\rho \leq w)$ and $\lambda = \log(1/q)$. As $w \downarrow 0$, with $u = \log(1/w)$,*

$$\log F_\rho(w) = -\frac{u^2}{2\lambda} + O(u \log u). \quad (10.1)$$

For every fixed $r \geq 0$ the same leading estimate holds for the kernel:

$$\log h_r(w) = -\frac{u^2}{2\lambda} + O(u \log u). \quad (10.2)$$

Proof. Write $b_j = \rho q^j$. If $T_\rho \leq w$, then every term with $b_j \geq w$ must satisfy $U_j \leq w/b_j$. There are $N_+ = u/\lambda + O(1)$ such terms. Independence gives

$$\log F_\rho(w) \leq \sum_{j=1}^{N_+} \log(w/b_j) = -\frac{u^2}{2\lambda} + O(u).$$

For the other direction, choose N to be the smallest integer such that the deterministic remaining support $\rho q^{N+1}/(1-q)$ is at most $w/2$. Again $N = u/\lambda + O(1)$. Impose, for $1 \leq j \leq N$,

$$U_j \leq \frac{w}{2N\rho q^j}.$$

For sufficiently small w all these thresholds lie below one: minimality of N gives $\rho q^N > (1-q)w/2$, so even the largest threshold is less than $1/[N(1-q)]$. On the displayed event the first N terms sum to at most $w/2$, and the remaining tail is at most $w/2$ deterministically. Thus

$$\begin{aligned} \log F_\rho(w) &\geq N \log w - N \log(2N\rho) + \lambda N(N+1)/2 \\ &= -\frac{u^2}{2\lambda} + O(u \log u). \end{aligned}$$

Together the two inequalities prove (10.1).

For $r \geq 1$, the positive-part expectation satisfies

$$(w/2)^r F_\rho(w/2) \leq \mathbb{E}(w - T_\rho)_+^r \leq w^r F_\rho(w).$$

For $r = 0$ use $h_0 = K e^{-w} F_\rho(w)$. Taking logarithms and applying (10.1) proves (10.2); a fixed multiplication of w changes u only by $O(1)$, and the additional factor w^r contributes only $O(u)$. $\square$

Proposition 10.2 (Size of the endpoint defect). *For fixed r and any fixed cutoff $D > 0$, define $\delta_r(z) = \int_0^D (1 - h_r(w)/z)_+ dw$. As $z \downarrow 0$,*

$$\log \delta_r(z) = -\sqrt{2\lambda \log(1/z)} + o\left(\sqrt{\log(1/z)}\right). \quad (10.3)$$

In particular, for every fixed $M > 0$, $\delta_r(c_n) = o(\Lambda_n^{-M})$.

Proof. Put $L = \log(1/z)$ and fix $\varepsilon \in (0, 1)$. Set

$$w_- = \exp\{-(1+\varepsilon)\sqrt{2\lambda L}\}, \quad w_+ = \exp\{-(1-\varepsilon)\sqrt{2\lambda L}\}.$$

The upper bound for the positive-part expectation used above, with monotonicity of F_ρ , implies

$$\sup_{0 < w \leq w_-} \frac{h_r(w)}{z} \rightarrow 0.$$

Indeed its logarithm is at most $L - (1+\varepsilon)^2 L + O(\sqrt{L} \log L)$, up to lower-order terms. Consequently $\delta_r(z) \geq w_-(1 - o(1))$.

Similarly, on $w_+ \leq w \leq D$, the lower bound

$$h_r(w) \geq \frac{K e^{-D}}{r!} (w_+/2)^r F_\rho(w_+/2)$$

for $r \geq 1$ tends to be much larger than z ; its logarithm after division by z is $L - (1-\varepsilon)^2 L + O(\sqrt{L} \log L) \rightarrow +\infty$. For $r = 0$, replace the positive-part factor by $K e^{-D} F_\rho(w_+)$. Hence the integrand vanishes throughout $[w_+, D]$ for small enough z , and $\delta_r(z) \leq w_+$. These bounds give

$$-(1+\varepsilon)\sqrt{2\lambda L} + o(1) \leq \log \delta_r(z) \leq -(1-\varepsilon)\sqrt{2\lambda L}.$$

Let $\varepsilon \downarrow 0$. The comparison with all inverse powers follows because $\sqrt{\Lambda_n}/\log \Lambda_n \rightarrow \infty$. $\square$

Proof of theorem 3.4. For $r = 0$, $h_0(w) = Ke^{-w}$ on $[C, \infty)$. If $z < Ke^{-C}$, the unique upper crossing is $b_0(z) = \log(K/z) > C$. Directly splitting the clipped-density integral at C and $b_0(z)$ gives the *exact* identity

$$J_0(z) = \log(K/z) + 1 - \int_0^C (1 - h_0(w)/z)_+ dw. \quad (10.4)$$

Combining with (3.6) proves the first part of (3.11). Since $\log(1/c_n) = \frac{1}{2} \log n + O(1)$, theorem 10.2 gives the second part.

Furthermore

$$\frac{(\log n)^3/n}{\delta_0(c_n)} \rightarrow 0 :$$

its logarithm is $-\log n + O(\sqrt{\log n}) + 3 \log \log n$. Thus $\mathcal{E}_n = \delta_0(c_n)(1 + o(1)) > 0$ eventually. Taking logarithms proves (3.12). $\square$

10.1 Why a scalar endpoint estimate reappears here

The conditional deficit distribution is not itself concentrated at zero: it converges to $R_\rho + \text{Exp}(1)$ when $r = 0$. Nevertheless the *overlap* clips that density at the shrinking level c_n . Its lower crossing probes exactly where $h_0(w)$ becomes comparable to $n^{-1/2}$. Inverting the log-squared endpoint estimate therefore produces $w \approx \exp(-\sqrt{\lambda \log n})$. This is an inversion of scales, not an invocation of a formal transseries without a remainder estimate. The sandwich proof above controls the logarithm of the correction without claiming a sharper prefactor.

11 Exact examples and a finite-simplex benchmark

11.1 The dyadic boundary polynomials

For $q = 1/2$ and $\rho = 1$, $C = 1$ and (4.3) gives

$$m_0^* = 1, \quad m_1^* = \frac{1}{2}, \quad m_2^* = \frac{5}{18}, \quad m_3^* = \frac{1}{6}.$$

Accordingly, for $w \geq 1$,

$$\begin{aligned} h_0(w) &= Ke^{-w}, \\ h_1(w) &= Ke^{-w}(w - \tfrac{1}{2}), \\ h_2(w) &= \tfrac{K}{2}e^{-w}(w^2 - w + \tfrac{5}{18}), \\ h_3(w) &= \tfrac{K}{6}e^{-w}(w^3 - \tfrac{3}{2}w^2 + \tfrac{5}{6}w - \tfrac{1}{6}). \end{aligned} \quad (11.1)$$

Numerically, the accompanying calculation gives

$$K \approx 1.626018147974, \quad \log K \approx 0.486134172179, \quad \kappa \approx -0.500013542859. \quad (11.2)$$

The conditioning threshold is therefore $x_n = 2^{-n}(n + \kappa + o(1))$ at this phase. These decimal values are floating-point evaluations, not interval-certified enclosures.

For $r = 0$, the three-term scaled overlap is $\Lambda_n + 1.486134172179 \dots$; its discrepancy has the eventual negative sign in (3.11). For $r = 1$, the first inverse-log correction is

$$\frac{\log \Lambda_n + \log K + 1/2}{\Lambda_n}.$$

At modest dimensions this term and the endpoint defect can have comparable sizes. Consequently the residual of the three-term formula need not be monotone in n .

11.2 The finite simplex: a classical consistency check

Consider instead the uniform law on

$$\{(y_1, \dots, y_n) : y_i \geq 0, \sum y_i \leq n\},$$

and compare its first $N = n - r$ coordinates with N independent rate-one exponentials. There is no infinite tail in this benchmark. The role of h_r is the gamma density

$$h_r^{\text{simp}}(w) = \frac{e^{-w} w^r}{r!},$$

and the exact normalizer is

$$Z_n^{\text{simp}} = e^{-n} \frac{n^n}{n!}. \quad (11.3)$$

Indeed, the numerator density of the observed sum V is

$$\frac{e^{-v} v^{N-1}}{(N-1)!} \frac{e^{-(n-v)} (n-v)^r}{r!} \mathbf{1}_{0 < v < n}.$$

Its integral is (11.3) by the beta integral, and $V/n \sim \text{Beta}(N, r+1)$ after normalization. In this case the logarithmic overlap expansion has $K = 1$ and no geometric phase.

The exact relative entropy is

$$\begin{aligned} D_1^{\text{simp}} = & -\log Z_n^{\text{simp}} - \frac{n(r+1)}{n+1} \\ & + r\{\log n + \psi(r+1) - \psi(n+1)\} - \log(r!), \end{aligned} \quad (11.4)$$

where ψ is the logarithmic derivative of the gamma function. This uses $\mathbb{E}(n - V) = n(r+1)/(n+1)$ and the beta log-moment. For integer parameters the latter also follows by differentiating the beta integral before setting its parameter to $r+1$. The limiting gamma differential entropy is

$$\mathfrak{h}_1(h_r^{\text{simp}}) = r+1 + \log(r!) - r\psi(r+1).$$

For general positive order,

$$\int_0^\infty (h_r^{\text{simp}}(w))^\alpha dw = \frac{\Gamma(\alpha r + 1)}{(r!)^\alpha \alpha^{\alpha r + 1}}. \quad (11.5)$$

These familiar gamma and simplex formulas are not originality claims. They test the factorial, the extra slack shape, the 2π normalization, and the sign of the entropy constant independently of the Fabius boundary calculation.

12 Reproducible diagnostics and their limitations

The directory accompanying this article contains one executable Python script, the generated CSV data and table fragments, three figures, and a machine-readable verification receipt. The proofs establish the asymptotic results. The computations check the algebra, constants, indexing, and finite-dimensional behavior; they are not substitutes for the proofs.

12.1 Exact algebra checks

The delivered run passed 228 exact symbolic assertions, with no failures. For six rational pairs (q, ρ) , moments through degree twelve computed from (4.3) were compared with an independent cumulant calculation. For $k \geq 2$ the cumulants used there are

$$\kappa_k(T_\rho) = \frac{B_k (\rho q)^k}{k (1 - q^k)},$$

with the first cumulant equal to $\rho q/[2(1 - q)]$. This formula follows by expanding the logarithm of the product of uniform moment-generating functions; equivalently, expand $\log((e^z - 1)/z)$ and sum the resulting geometric series. The standard moment-cumulant recursion then gives a second exact route to the moments. This produced 72 assertions.

The differentiated tail-polynomial identities were checked through $r = 12$ for the same six parameter pairs, producing another 72 assertions. Finally 84 finite-simplex polynomial integrals were evaluated exactly for $1 \leq N \leq 12$, $0 \leq r \leq 6$. Each integral was compared with

$$\int_0^{N+r} v^{N-1} (N + r - v)^r dv = \frac{(N + r)^{N+r} (N - 1)! r!}{(N + r)!}.$$

These checks verify finite identities in exact rational arithmetic. They do not verify the infinite-series analysis or the limiting arguments in a proof assistant.

12.2 Positive boundary convolution and a signed gamma mixture

For the dyadic case $q = 1/2$, $\rho = 1$, the original boundary variable T_ρ was approximated by convolving midpoint discretizations of its uniform summands on a dyadic grid. The unrevealed tail was replaced by its exact mean. Repeated numerical integration of the resulting CDF gives $E(w - T)_+^r / r!$ for $0 < w < 1$; above one the exact moment polynomials (11.1) were used. Comparing grid sizes 2^{16} and 2^{17} gave a maximum sampled kernel difference of approximately 7.20×10^{-10} . This is a refinement diagnostic, not a rigorous uniform error bound.

The bulk density was evaluated using an inclusion-exclusion sum of shifted gamma densities. If a collection of caps a_i is retained, the formula is

$$f(v) = \frac{1}{\prod_i (1 - e^{-a_i})} \sum_J (-1)^{|J|} e^{-a_J} \gamma_N(v - a_J), \quad a_J = \sum_{j \in J} a_j, \quad (12.1)$$

where $N = n - r$ and γ_N is zero at nonpositive arguments. This follows by translating each violated exponential coordinate by its cap before integrating on the sum. The calculation retained the caps 1, 2, 4, 8, 16, 32 and removed all caps at least 64. The union probability of those removed upper-tail events is less than 2×10^{-28} under the exponential reference. This event bound does not control floating-point cancellation in (12.1); no interval arithmetic was used.

There were 24 geometric cases: $r = 0, 1, 2, 3$ at $n = 64, 256, 1024, 4096, 16384, 65536$. The normalizer, overlap, and relative entropy were integrated from their one-dimensional formulas. An additional 24 finite-simplex evaluations tested the order-zero transition. All numerical tables are generated by the supplied script.

n	r	$\sqrt{2\pi n} \mathcal{O}_{n,r}$	Prediction	Residual
64	0	4.284589	4.484514	-0.199925
1,024	0	5.735118	5.870809	-0.135690
65,536	0	7.865122	7.950250	-0.085128
64	1	5.790207	5.582586	+0.207621
1,024	1	7.578840	7.348924	+0.229916
65,536	1	10.045763	9.816516	+0.229247
64	2	6.964975	5.987511	+0.977464
1,024	2	9.074007	8.133892	+0.940114
65,536	2	11.846601	10.989635	+0.856965
64	3	7.919538	5.986971	+1.932567
1,024	3	10.354197	8.513395	+1.840802
65,536	3	13.426689	11.757289	+1.669400

Table 1: Dyadic geometric calculations. “Prediction” is $\Lambda_n + r \log \Lambda_n + \log K - \log(r!) + 1$. The residuals show the very slow convergence of this truncated asymptotic expansion, especially when r is positive. Decimal entries are diagnostics, not certified digits.

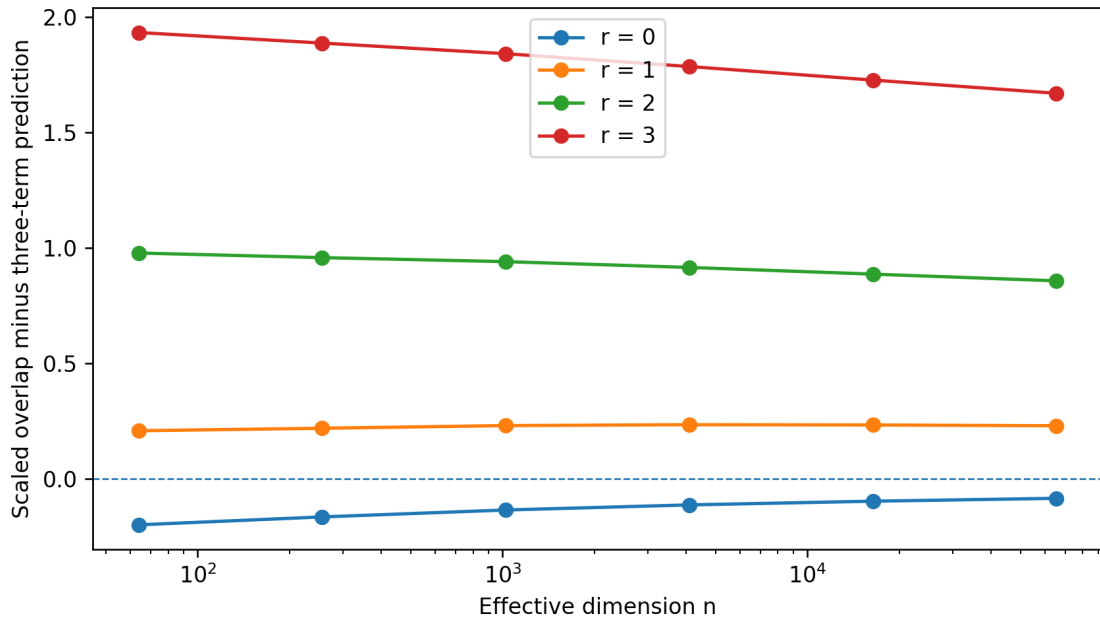


Figure 1: Scaled overlap minus the three-term prediction in [theorem 3.1](#). The theorem is a fixed- r asymptotic statement; large visible residuals at these dimensions do not contradict it. The explicit kernel reduction retains those corrections.

n	r	Scaled overlap	$J_r(c_n)$	$\delta_r(c_n)$	Difference
1,024	0	5.735118	5.744889	0.125920	-0.009771
65,536	0	7.865122	7.865720	0.084530	-0.000598
1,024	1	7.578840	7.596856	0.259043	-0.018016
65,536	1	10.045763	10.046779	0.171746	-0.001016
1,024	2	9.074007	9.100505	0.471064	-0.026498
65,536	2	11.846601	11.848033	0.312294	-0.001433
1,024	3	10.354197	10.389753	0.767447	-0.035556
65,536	3	13.426689	13.428555	0.513576	-0.001866

Table 2: The fixed-kernel reduction is considerably more accurate at finite n than the three-term expansion. “Difference” is the scaled overlap minus $J_r(c_n)$. The endpoint defect remains appreciable even at $n = 65536$, particularly for larger fixed r .

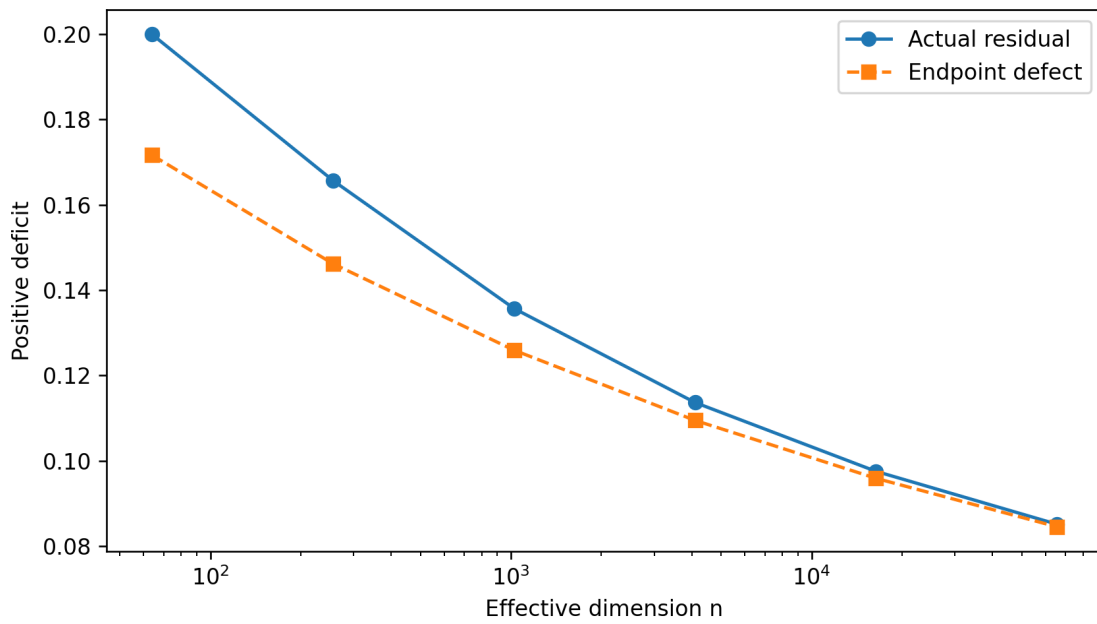


Figure 2: For $r = 0$, the deficit from $\Lambda_n + \log K + 1$ compared with the boundary defect $\delta_0(c_n)$. Their difference is the bulk local-density correction. This plot checks the structural identity; it is not a numerical determination of the eventual logarithmic exponent in [theorem 3.4](#).

r	$h(H_r)$	$h_{1/2}(H_r)$	Normalization error
0	1.14084719	1.48847380	-8.79e-11
1	1.60461229	1.86924365	-7.28e-11
2	1.85877132	2.09397602	-2.08e-11
3	2.02988383	2.25179053	-1.72e-12

Table 3: Boundary entropy constants for $q = 1/2$, $\rho = 1$. The symbols $h(H_r)$ and $h_{1/2}(H_r)$ in the table denote $\mathfrak{h}_1(h_r)$ and $\mathfrak{h}_{1/2}(h_r)$, respectively. The last column is the numerical integral of the kernel minus one, not an a priori error estimate.

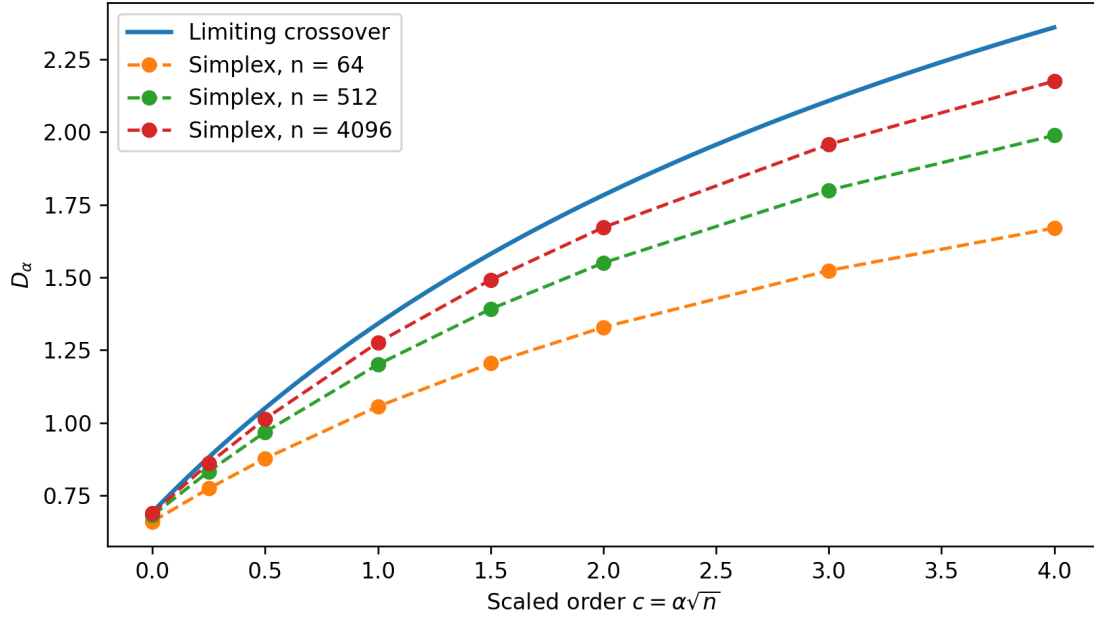


Figure 3: The universal order-zero transition, compared with finite-simplex integrals at $r = 0$. These are an independent benchmark, not geometric-series observations. The finite-dimensional bias is visible, especially for larger scaled order c .

12.3 How to reproduce the calculations

Run `python code/verify.py` from the package directory. The verification receipt is written to `data/verification.json`. The script also writes exact algebra records, CSV diagnostics, table fragments, and the figures. The delivered numerical run used Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0, SymPy 1.14.0, and mpmath 1.3.0. The environment is recorded in the receipt. The PDF can be rebuilt independently of the numerical calculation with `latexmk -pdf article.tex`; the generated figures and table fragments are already included.

13 Further research questions and a development plan

The results settle the fixed- r , bounded-complement slice of the repository’s critical-complement question. They leave a structured set of next problems. The statements in this section are questions and proposed targets, not additional theorems.

13.1 From fixed complements to growing complements

Question 1 (A uniform mesoscopic complement theory). Let $r = r_n \rightarrow \infty$ while $r_n = o(n)$, and observe $\{r_n + 1, \dots, n\}$. What uniform overlap and information formulas interpolate between the non-Gaussian critical kernel and the fixed-fraction Gaussian profile?

The early-cap cancellation remains exact whenever the deleted caps exceed the constraint, but the gamma shape in H_{r_n} now grows. The polynomial degree, entropy, and clipping threshold all move. The main obstruction is no longer discovering a kernel; it is controlling the bulk density on the entire range where that kernel is clipped. A useful first target is an explicit regime such as $r_n(\log n)^2 = o(n)$, followed by identification of the sharp condition. That displayed regime is a suggested starting point, not a proved sufficient condition here.

Question 2 (The optimal finite- n error). Can the $O((\log n)^3/n)$ error in (3.6) be replaced by an explicit signed expansion with a computable remainder, uniformly over a compact set of phases?

The proof deliberately takes suprema over a logarithmic window. Retaining the quadratic and cubic terms of the centered gamma core and the exact low cumulants of the short boundary block should give a more efficient expansion. Its coefficients would interact with truncated moments of $\min(1, h_r/c_n)$, rather than only ordinary moments. A rigorous answer must control the signed gamma mixture or another positive representation without assuming a formal expansion is uniformly integrable.

13.2 Endpoint defects beyond their leading logarithm

Question 3 (A sharp asymptotic equivalent for the defect). Determine $\delta_r(c_n)$ up to a multiplicative $1 + o(1)$ error, including any phase-sensitive corrections inherited from the small-ball asymptotics of T_ρ .

We proved its logarithm only to leading square-root order. A refined small-ball formula has to be inverted at the moving level $h_r(w) = c_n$, and then integrated across the transition. The integral need not equal the lower crossing to relative error $o(1)$ without an additional steepness estimate. This is where the repository's exact-phase scalar endpoint machinery could supply more than the elementary bounds used in this article.

Question 4 (Inverse reconstruction from critical information). To what extent do the functions $r \mapsto \mathfrak{h}_\alpha(h_r)$, the phase constant K , and the defects $\delta_r(z)$ determine the underlying boundary law?

The upper-tail polynomials already recover successive ordinary moments once K is known. Since the boundary support is compact, all moments determine its law. The more difficult question is how much information is retained when only entropies or asymptotic coefficients of a fixed observed block are available. A stability result would be more useful than uniqueness alone.

13.3 Information-order transitions and different constraints

Question 5 (Uniform matching of Rényi orders). Find a single approximation valid from $\alpha = 0$ through $\alpha \sqrt{n} \rightarrow \infty$, and into fixed positive orders.

The limiting function in (9.4) matches the formal small-order limit of the fixed-order formula. A proof must control $h_r(w)^\alpha$ when its mass explores both the $w = O(1)$ and $w = O(\sqrt{n})$ regions. The first correction may separate universal bulk terms from r - and phase-dependent terms. Uniform treatment near order one would then connect this matching to relative entropy without a removable-singularity loss.

Question 6 (Several inequalities or a thin conditioning window). What replaces the single gamma-slack kernel under two independent linear constraints, a cone constraint, or conditioning on $\mu_n - a_n \leq t_n S_q \leq \mu_n$?

For a fixed-width window the one-sided exponential slack must be truncated. For several constraints the relevant local object is a multivariate boundary law integrated over a feasible cone, and the canonical normalizing density is matrix-valued. One should first prove the analogue of [theorem 5.1](#) in a concrete two-dimensional example; simply replacing a scalar variance by a determinant does not account for the cone geometry.

13.4 Other coordinate laws and formal computation

Question 7 (Gamma universality and its failures). Suppose the coordinate density behaves near zero like $u^{\beta-1} \ell(u)$, with $\beta > 0$. Under what hypotheses does the critical kernel become a boundary tail plus a gamma variable of shape $r\beta + 1$?

A gamma limit for each strongly tilted lost coordinate is a natural candidate. Unlike the uniform case, however, an eventually exact cancellation of all early caps and density factors is generally unavailable. Rates for the local density and for the approximation of the early digits must be proved jointly. Slowly varying factors, nonlog-concave densities, and atoms at zero may yield genuinely different behavior.

Question 8 (Formal verification and certified numerical kernels). Can the moment-polynomial identities, exact kernel reduction, and local gamma-core bounds be formalized as reusable Lean lemmas, and can the accompanying numerical formulas be evaluated with certified intervals?

A practical dependency order is: finite product tilting and pushforward likelihoods; gamma convolution and polynomial tails; exact rational moment recursion; the gamma-core local estimate; normalization and clipped-density comparison; finally entropy and moving-order limits. The finite algebra is a comparatively small first formalization target. The analytic layer needs countable product probability spaces, measure changes, convolution, and dominated-convergence infrastructure. No uncompiled Lean skeleton is included as evidence of verification.

For certified computation, a positive boundary-convolution algorithm with interval CDF bounds avoids alternating cancellation near the flat endpoint. The bulk can then be bounded by interval versions of the gamma-core decomposition. A final error budget must separately account for tail truncation, discretization, quadrature, root location, and rounding; the current refinement comparisons do not provide that budget.

14 Conclusion and proof-status boundary

The earlier variance-fraction theorem says that an almost complete active block eventually separates from its tilted product marginal. In the fixed critical-complement regime, the present results quantify that separation and identify what survives below the Gaussian scale. The missing early digits contribute a gamma shape, the infinite geometric tail contributes a phase-dependent density, and clipping that density determines the exact logarithmic overlap hierarchy.

At every fixed positive information order, a boundary differential entropy supplies the finite correction to the diverging bulk term. At order zero the support remains macroscopic, producing $\log 2$ instead. The universal crossover makes the noncommutation of these limits explicit. Finally, when no early digit is omitted, all inverse-logarithmic upper-tail corrections disappear; the remaining signed discrepancy directly reveals the flat Fabius endpoint.

The mathematical claims labelled as theorems and propositions have proofs in this manuscript. They have not been externally refereed or compiled in Lean. The numerical script verifies finite algebraic identities and performs floating-point diagnostics only. The proposed novelty is relative to the inspected repository question and the reviewed references; neither an exhaustive priority search nor a resolution of every formulation of the critical-complement problem is claimed.

A Repository provenance and inspection scope

The repository tree was pinned at

b8b0fa2184a044d46ce9ed0f25f88d7bb60fa042

The baseline manuscript is located under

Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/
representations/Sharp_Conditioning_Laws_Uniform_Random_Series/

Its `article.tex` blob identifier is

55945402782d507eb59dfd195514037ef41fb964.

We inspected the baseline README and claim-status ledger, and the source’s explicit research questions, including `q:critical` and `q:edgeworth`, at this pinned reference. We also inspected its later proof/discussion sections for context. The full repository was not rebuilt, nor were all archived manuscripts exhaustively searched. GitHub code search may lag a branch tip, so absence from a search result was not treated as proof of novelty.

The baseline’s claim ledger explicitly excludes quantitative uniform bounds near retained variance fraction one. Our error estimates are for fixed r and fixed geometric parameters, not the uniform shrinking-complement-fraction bounds posed in its most general question. This distinction is part of the theorem statements, not merely an implementation limitation.

The external review used the primary arXiv records of the Fabius, small-deviation, and Rényi references, together with the original conditional-limit reference’s bibliographic record. This was a targeted relevance and attribution check, not an exhaustive full-text literature survey. All analytic steps specific to the new theorem package are supplied in the article.

B Package map

`article.tex` and `article.pdf` are the complete source and compiled manuscript. The source inputs the generated table fragments from `data/` and the three PNG figures in `figures/`. `code/verify.py` reproduces the exact checks, diagnostics, tables, and figures. `data/exact_algebra.json` contains the rational moments and tail polynomials; `data/verification.json` records counts, numerical constants, and the executed environment. `README.md`, `CLAIM_STATUS.md`, `provenance.json`, and `validation.json` document the scope and artifact checks. No font files are distributed.

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